

Book Recommendation System

Mini Project Report

Submitted in partial fulfillment of the requirements

For the degree of

Bachelor of Engineering (Computer Engineering)

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Internal Approval Sheet



TERNA ENGINEERING COLLEGE, NERUL

Department of Computer Engineering

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CERTIFICATE

This is to certify that project entitled “Book Recommendation System” is
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Submitted to the University of Mumbai in partial fulfilment of the requirement for
the award of the Bachelor of Engineering (Computer Engineering).

Guide

Head of Department

Principal

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Chapter 1

Introduction

1.1 Motivation :

As the digital age progresses, the sheer volume of content available to users - books, music, movies has grown exponentially. Consequently, recommendation systems have become a critical aspect of web applications, assisting users in finding the most relevant and personalized content.

The Book Recommendation System is designed to simplify the process of book discovery by providing personalized recommendations to users based on their past interactions or the popularity of books within the system. This project focuses on using Machine Learning (ML) & Natural Language Processing (NLP) techniques in conjunction with machine learning algorithms to create a highly effective recommendation model.

1.2 Problem Statement:

Readers often find it difficult to select new books due to the overwhelming volume of choices available. This project addresses the challenge by implementing a Book Recommendation System that employs two main strategies:

1. Popularity-Based Recommendations: Highlights top-rated books based on user feedback.
2. Collaborative Filtering: Recommends books based on a user's interaction with other similar books and the preferences of other users.

1.3 Objective

The main objective of this project is to:

1. Develop a web-based system that allows users to search for books and receive personalized recommendations.
2. Implement two different recommendation techniques popularity-based and collaborative filtering to cater to different user preferences.
3. Create a user-friendly Graphical User Interface (GUI) that displays the top 50 books and provides recommendations in real-time.

1.4 Scope of the Project:

This project is designed as a scalable web application capable of handling a growing user base and dataset. The system supports both popularity-based and personalized recommendations using collaborative filtering techniques. It is built using Flask for the backend, Bootstrap for the frontend, and Python libraries for machine learning.

Chapter 2

Software & Hardware Requirements

1. **Operating System:** Windows 10/11 or Linux (Ubuntu 20.04 or above).
2. **Programming Language:** Python 3.x (latest version recommended).
3. **Integrated Development Environment (IDE):** Visual Studio Code or PyCharm.
4. **Web Framework:** Flask (micro web framework written in Python).
5. **Libraries Used:**
 - 5.1 **Flask:** To manage the web server and routing.
 - 5.2 **Numpy:** For handling arrays and mathematical operations.
 - 5.3 **Pandas:** For managing and processing data in tabular form.
 - 5.4 **Pickle:** For serializing and de-serializing Python objects (used to store the model).
 - 5.5 **Scikit-learn:** For implementing the machine learning algorithms (Collaborative Filtering).
6. **Frontend Technologies:** HTML5, CSS3, Bootstrap 3.3.7.
7. **Databases:** CSV files containing books, users, and ratings data.

Chapter 3

Software Design

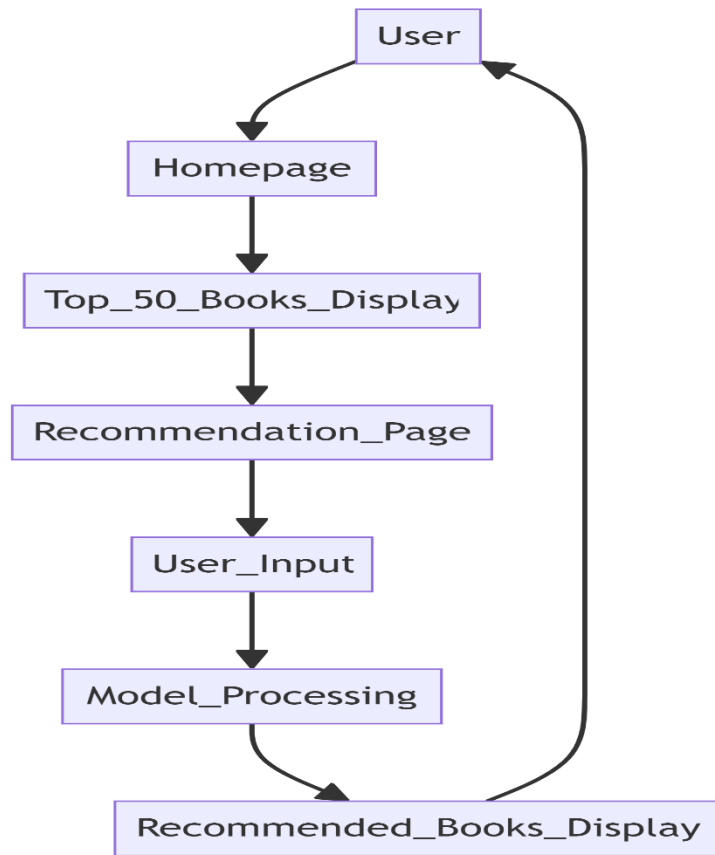


Fig: 3.1

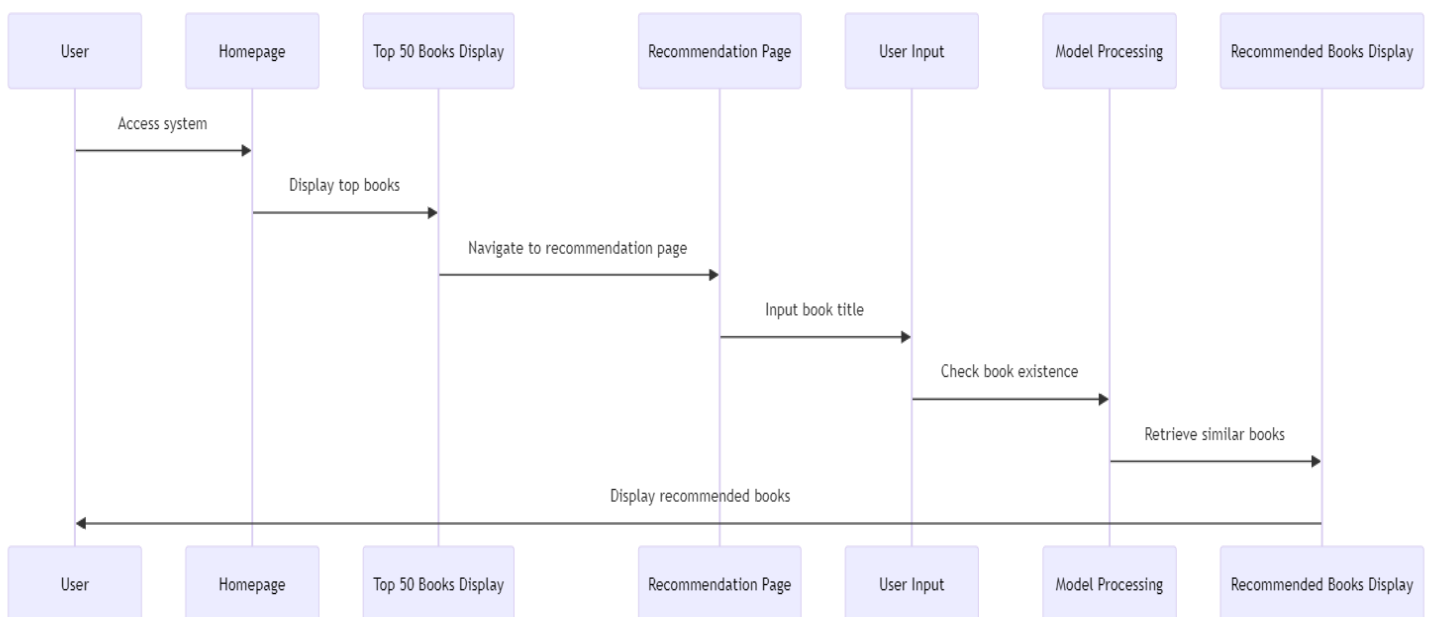


Fig: 3.2

Chapter 4

Algorithms & Methods Used

4.1 Popularity-Based Recommender System:

This method ranks books based on their popularity metrics, specifically the **number of ratings** and the **average rating** for each book. The implementation follows these steps:

1. **Data Aggregation:** Ratings data is aggregated to count the number of ratings and calculate the average rating for each book.
2. **Filtering:** Only books with a minimum of 250 ratings are considered.
3. **Top 50 Selection:** The books are then sorted based on their average rating, and the top 50 books are selected for display.

Algorithmic Flow:

1. Aggregate rating data.
2. Filter out books with fewer than 250 ratings.
3. Sort books by average rating.
4. Select the top 50 books for display.

4.2 Collaborative Filtering Based Recommender System:

Collaborative filtering is a method used to recommend items (books) to a user by finding similarities between items based on user ratings. The **cosine similarity** method is used to calculate the similarity between books based on user ratings.

Steps in Collaborative Filtering:

1. **Pivot Table Creation:** A pivot table is created where rows represent book titles, columns represent user IDs, and cells contain the rating a user has given to a book.
2. **Cosine Similarity Calculation:** Using the **cosine similarity** metric, the similarity between books is calculated.
3. **Recommendation Retrieval:** When a user inputs a book, the system identifies similar books based on their similarity scores and presents them as recommendations.

Algorithmic Flow:

1. Create a pivot table using books and users.
2. Compute cosine similarity between books.
3. Identify top similar books based on the input book.
4. Present recommendations.

Chapter 5

Project Implementation

5.1 Data Collection and Preprocessing:

Datasets Used:

1. **Books Dataset:** Contains details about books including title, author, and image URL.
2. **Users Dataset:** Contains information about users, primarily used for collaborative filtering.
3. **Ratings Dataset:** Contains user ratings for different books. This dataset forms the basis for both the popularity-based and collaborative filtering models.

Preprocessing Steps:

1. **Data Cleaning:** Handle missing values and remove any duplicates in the data.
2. **Merging Datasets:** Combine the books and ratings data into a single dataset that can be used for modeling.
3. **Data Splitting:** Split the data into training and test sets for evaluation (optional if further model training is needed).

5.2 Model Training and Serialization:

The models are trained using the pre-processed datasets:

1. **Popularity Model:** This model is trained by calculating the number of ratings and the average rating for each book. The top 50 books are then serialized using **Pickle** for fast access during runtime.
2. **Collaborative Filtering Model:** The pivot table is created, and cosine similarity scores are calculated for every book. The similarity matrix is then serialized for efficient retrieval of recommendations.

Pickle Files:

1. **popular.pkl:** Contains the top 50 popular books.
2. **similarity_scores.pkl:** Stores the cosine similarity scores between books.
3. **pt.pkl:** Contains the pivot table used for collaborative filtering.

5.3 Flask Application:

The **Flask App** serves as the backbone of the web application, handling user interactions and routing requests:

1. **Homepage Route (/):** Displays the top 50 books using the popularity-based model.
2. **Recommendation Page Route (/recommend):** Handles the input of book names and retrieves recommendations from the collaborative filtering model.

Chapter 6

Program

Input Code:

App.py:

```
app.py X
book-recommender-system > app.py > recommend
1 from flask import Flask,render_template,request
2 import pickle
3 import numpy as np
4
5 popular_df = pickle.load(open('popular.pkl','rb'))
6 pt = pickle.load(open('pt.pkl','rb'))
7 books = pickle.load(open('books.pkl','rb'))
8 similarity_scores = pickle.load(open('similarity_scores.pkl','rb'))
9
10 app = Flask(__name__)
11
12 @app.route('/')
13 def index():
14     return render_template('index.html',
15                             book_name = list(popular_df['Book-Title'].values),
16                             author=list(popular_df['Book-Author'].values),
17                             image=list(popular_df['Image-URL-M'].values),
18                             votes=list(popular_df['num_ratings'].values),
19                             rating=list(popular_df['avg_rating'].values)
20     )
21
22 @app.route('/recommend')
23 def recommend_ui():
24     return render_template('recommend.html')
25
26 @app.route('/recommend_books',methods=['post'])
27 def recommend():
28     user_input = request.form.get('user_input')
29
30     # Check if user_input is empty
31     if not user_input:
32         return render_template('recommend.html', data=[], error_message="No results found or please enter a valid book name.")
33
34     # Try to find the index of the book title
35     try:
36         index = np.where(pt.index == user_input)[0][0]
37     except IndexError:
```

```
app.py X
book-recommender-system > app.py > recommend
27 def recommend():
35     try:
36         index = np.where(pt.index == user_input)[0][0]
37     except IndexError:
38         # If the book title is not found, display an error message
39         return render_template('recommend.html', data=[], error_message="No results found or please enter a valid book name.")
40
41     similar_items = sorted(list(enumerate(similarity_scores[index])), key=lambda x: x[1], reverse=True)[1:5]
42
43     data = []
44     for i in similar_items:
45         item = []
46         temp_df = books[books['Book-Title'] == pt.index[i[0]]]
47         item.extend(list(temp_df.drop_duplicates('Book-Title')['Book-Title'].values))
48         item.extend(list(temp_df.drop_duplicates('Book-Title')['Book-Author'].values))
49         item.extend(list(temp_df.drop_duplicates('Book-Title')['Image-URL-M'].values))
50
51         data.append(item)
52
53     print(data)
54
55     return render_template('recommend.html',data=data)
56
57 if __name__ == '__main__':
58     app.run(debug=True)
```

Index.html :

```
index.html X
book-recommender-system > templates > index.html > html > head
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Book Recommender System</title>
7      <!-- Latest compiled and minified CSS -->
8      <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap@3.3.7/dist/css/bootstrap.min.css"
9            integrity="sha384-BVYiSIFeK1dGmJRAkycuHAHRg320mUcww7on3RYdg4Va+PmSTsz/K68vbdEjh4u" crossorigin="anonymous">
10     <!-- Custom CSS -->
11     <style>
12         body {
13             background-color: #f4f4f9;
14             font-family: 'Arial', sans-serif;
15             margin-top: 60px; /* Adjusted margin */
16         }
17
18
19         .navbar {
20             background-color: #007bff;
21             position: fixed; /* Makes the navbar fixed */
22             top: 0; /* Aligns it to the top */
23             left: 0; /* Aligns it to the left */
24             right: 0; /* Ensures it covers the full width */
25             z-index: 1000; /* Ensures it stays above other content */
26         }
27
28         .navbar-brand, .navbar-nav li a {
29             color: white !important;
30             font-size: 1.8rem;
31             font-weight: bold;
32         }
33
34         h1 {
35             color: #343a40;
36             font-weight: 700;
```

```
index.html X
book-recommender-system > templates > index.html > html > head
2  <html lang="en">
3  <head>
11     <style>
34         h1 {
35             color: #343a40;
36             font-weight: 700;
37             text-align: center;
38             margin-top: 40px;
39             font-size: 2.5rem;
40         }
41
42         .book-card {
43             background-color: #fff;
44             border-radius: 10px;
45             box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
46             padding: 20px;
47             margin-bottom: 30px;
48             text-align: center;
49             height: 350px; /* Fixed height for uniformity */
50             display: flex;
51             flex-direction: column;
52             justify-content: space-between;
53         }
54
55         .book-card img {
56             max-width: 100%;
57             height: 200px; /* Fixed image height */
58             object-fit: cover; /* Ensure image fills the space */
59             border-radius: 5px;
60         }
61
62         .book-card h4, .book-card p {
63             color: #007bff;
64             font-weight: bold;
65         }
66
```

```

index.html X
book-recommender-system > templates > index.html > html > body > nav.navbar.navbar-expand-lg > ul.nav.navbar-nav
 2 <html lang="en">
 3 <head>
11 <style>
65     }
66
67     .book-card h4 {
68         font-size: 1.2rem;
69     }
70
71     .book-card p {
72         font-size: 1rem;
73     }
74
75     .book-card .votes, .book-card .rating {
76         color: #6c757d;
77         font-size: 0.9rem;
78     }
79
80     .row {
81         display: flex;
82         flex-wrap: wrap;
83         gap: 20px; /* Ensure consistent spacing between cards */
84     }
85
86     .col-md-3 {
87         flex: 1 1 calc(25% - 20px); /* Ensures 4 columns with gaps */
88         box-sizing: border-box;
89     }
90 </style>
91 </head>
92 <body>
93
94     <nav class="navbar navbar-expand-lg">
95         <a class="navbar-brand" href="#">Book Recommender</a>
96         <ul class="nav navbar-nav">
97             <li><a href="/">Home</a></li>

```

```

index.html X
book-recommender-system > templates > index.html > html > body > nav.navbar.navbar-expand-lg > ul.nav.navbar-nav
 2 <html lang="en">
92 <body>
94     <nav class="navbar navbar-expand-lg">
96         <ul class="nav navbar-nav">
97             <li><a href="/">Home</a></li>
98             <li><a href="/recommend">Recommend</a></li>
99             <!-- <li><a href="#">Contact</a></li> -->
100         </ul>
101     </nav>
102
103
104     <!-- <nav class="navbar navbar-expand-lg">
105         <div class="container">
106             <a class="navbar-brand" href="#">Book Recommender</a>
107             <div class="collapse navbar-collapse">
108                 <ul class="nav navbar-nav navbar-right">
109                     <li><a href="/">Home</a></li>
110                     <li><a href="/recommend">Recommend</a></li>
111                 </ul>
112             </div>
113         </div>
114     </nav> -->
115
116
117     <div class="container">
118         <h1>Top 50 Books</h1>
119         <div class="row">
120             {% for i in range(book_name|length) %}
121                 <div class="col-md-3">
122                     <div class="book-card">
123                         
124                         <h4>{{ book_name[i] }}</h4>
125                         <p>{{ author[i] }}</p>
126                         <p class="votes">Votes: {{ votes[i] }}</p>
127                         <p class="rating">Rating: {{ rating[i] }}</p>
128                     </div>

```

```
index.html X
book-recommender-system > templates > index.html > html > body > div.container > div.row > div.col-md-3
2   <html lang="en">
92  <body>
117    <div class="container">
119      <div class="row">
121        <div class="col-md-3">
125          <p>{{ author[i] }}</p>
126          <p class="votes">Votes: {{ votes[i] }}</p>
127          <p class="rating">Rating: {{ rating[i] }}</p>
128        </div>
129      </div>
130    {% endfor %}
131  </div>
132 </div>
133
134 </body>
135 </html>
136
```

Recommend.html:

```
recommend.html 7 X
book-recommender-system > templates > recommend.html > html > head > style > .book-card
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8">
5    <meta name="viewport" content="width=device-width, initial-scale=1.0">
6    <title>Book Recommender System</title>
7    <!-- Latest compiled and minified CSS -->
8    <link rel="stylesheet"
9      href="https://cdn.jsdelivr.net/npm/bootstrap@3.3.7/dist/css/bootstrap.min.css"
10     integrity="sha384-BVYiiSIFeK1dGmJRAKycuHAHRg320mUcww7on3RYdg4Va+PmSTsz/K68vbdEjh4u"
11     crossorigin="anonymous">
12    <!-- Custom CSS -->
13    <style>
14      body {
15        background-color: #f4f4f9;
16        font-family: 'Arial', sans-serif;
17        margin-top: 60px; /* Adjusted margin */
18      }
19
20      .navbar {
21        background-color: #007bff;
22        position: fixed; /* Makes the navbar fixed */
23        top: 0; /* Aligns it to the top */
24        left: 0; /* Aligns it to the left */
25        right: 0; /* Ensures it covers the full width */
26        z-index: 1000; /* Keeps it on top */
27      }
28
29      .navbar-brand, .navbar-nav li a {
30        color: white !important;
31        font-size: 1.8rem;
32        font-weight: bold;
33      }
34
35      h1 {
36        color: #343a40;
37        font-weight: 700;
38      }
39    </style>
40  </head>
41  <body>
42    <div class="container">
43      <div class="row">
44        <div class="col-md-3">
45          <h1>Book Recommender System</h1>
46          <h2>Latest Books</h2>
47          <div class="list-group">
48            <div class="list-group-item">
49              <div class="row">
50                <div class="col-md-2">
51                  <img alt="Book cover image" data-bbox="100 100 200 200"/>
52                <div class="col-md-10">
53                  <div class="row">
54                    <div class="col-md-6">
55                      <h3>Book Title</h3>
56                      <p>Author Name</p>
57                    <div class="col-md-6">
58                      <p>Votes: {{ votes[i] }}</p>
59                      <p>Rating: {{ rating[i] }}</p>
60                    </div>
61                  </div>
62                  <p>{{ description }}</p>
63                </div>
64              </div>
65            </div>
66          </div>
67        </div>
68      </div>
69    </div>
70  </body>
71 </html>
```

```

recommend.html 7 X
book-recommender-system > templates > recommend.html > html > head > style
 2  <html lang="en">
 3  <head>
13  <style>
35      h1 {
36          color: #343a40;
37          font-weight: 700;
38          text-align: center;
39          margin-top: 40px;
40          font-size: 2.5rem;
41      }
42
43      .book-card {
44          background-color: #fff;
45          border-radius: 10px;
46          box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
47          padding: 20px;
48          margin-bottom: 30px;
49          text-align: center;
50          height: 350px; /* Fixed height for uniformity */
51          display: flex;
52          flex-direction: column;
53          justify-content: space-between;
54      }
55
56      .book-card img {
57          max-width: 100%;
58          height: 200px; /* Fixed image height */
59          object-fit: cover;
60          border-radius: 5px;
61      }
62
63      .book-card h4, .book-card p {
64          color: #007bff;
65          font-weight: bold;
66      }
67
68      .row {

```

```

recommend.html 7 X
book-recommender-system > templates > recommend.html > html > head > style
 2  <html lang="en">
 3  <head>
13  <style>
66      }
67
68      .row {
69          display: flex;
70          flex-wrap: wrap;
71          justify-content: space-between; /* Ensures cards take up space proportionally */
72      }
73
74      .col-md-3 {
75          flex: 1 1 calc(25% - 20px);
76          box-sizing: border-box;
77          margin-bottom: 20px;
78      }
79
80      /* Custom styles for form */
81      .form-container {
82          margin-top: 40px;
83          display: flex;
84          justify-content: center;
85          align-items: center; /* Center items vertically */
86      }
87
88      .search-bar {
89          display: flex;
90          align-items: center;
91          position: relative;
92          flex: 1; /* Makes the search bar expand to take available space */
93      }
94
95      .search-bar input {
96          padding-left: 40px;
97          height: 50px; /* Adjusted height */
98          border-radius: 25px; /* Rounded corners */
99          border: 2px solid #007bff; /* Border color */

```

```

recommend.html 7 X
book-recommender-system > templates > recommend.html > html > head > style > .book-card h4 > .book-card p
 2  <html lang="en">
 3  <head>
13    <style>
95      .search-bar input {
98        border-radius: 25px; /* Rounded corners */
99        border: 2px solid #007bff; /* Border color */
100        box-shadow: 0 2px 4px rgba(0, 0, 0, 0.1); /* Shadow effect */
101      }
102
103      .search-bar .search-icon {
104        position: absolute;
105        left: 10px;
106        top: 50%; /* Center vertically */
107        transform: translateY(-50%); /* Adjust position */
108        font-size: 18px;
109        color: #007bff;
110      }
111
112      .btn-clear, .btn-recommend {
113        border-radius: 25px; /* Rounded corners */
114        height: 50px; /* Adjusted height */
115        padding: 0 20px; /* Added horizontal padding for better size */
116        margin-left: 10px; /* Space between buttons */
117        box-shadow: 0 2px 4px rgba(0, 0, 0, 0.2); /* Shadow effect */
118      }
119
120      .btn-clear {
121        background-color: #dc3545;
122        color: white;
123      }
124
125      .btn-recommend {
126        background-color: #007bff;
127        color: white;
128      }
129
130      /* Error and popup messages */

```

```

recommend.html 7 X
book-recommender-system > templates > recommend.html > html > head > style > .book-card h4 > .book-card p
 2  <html lang="en">
 3  <head>
13    <style>
131      .alert {
132        margin-top: 20px;
133        display: none; /* Hidden by default */
134        text-align: center; /* Center the text */
135        width: 80%; /* Adjust width as needed */
136        margin-left: auto; /* Center horizontally */
137        margin-right: auto; /* Center horizontally */
138      }
139    </style>
140  </head>
141  <body>
142
143    <nav class="navbar navbar-expand-lg">
144      <a class="navbar-brand" href="#">Book Recommender</a>
145      <ul class="nav navbar-nav">
146        <li><a href="/">Home</a></li>
147        <li><a href="/recommend">Recommend</a></li>
148        <!-- <li><a href="#">Contact</a></li> -->
149      </ul>
150    </nav>
151
152    <div class="container">
153      <h1>Recommend Books</h1>
154      <div class="form-container">
155        <form id="searchForm" action="/recommend_books" method="post" class="form-inline"
156          style="display: flex; align-items: center;">
157          <div class="search-bar">
158            <span class="glyphicon glyphicon-search search-icon"></span>
159            <input id="bookInput" name="user_input" type="text" class="form-control"
160              placeholder="Enter a book name" value="{{ user_input }}">
161          </div>
162          <input type="submit" class="btn btn-recommend btn-lg" value="Get Recommendations">
163          <button type="button" id="clearBtn" class="btn btn-clear btn-lg">Clear</button>
164        </form>

```

```

recommend.html 7 x
book-recommender-system > templates > recommend.html > html > head > style > .book-card h4 > .book-card p
2 <html lang="en">
141 <body>
152 <div class="container">
154 <div class="form-container">
164 </form>
165 </div>
166
167 <!-- Error message for invalid or empty input -->
168 <!-- <div class="alert alert-danger" id="errorMessage">
169 <strong>NOTE:</strong> No results found or please enter a valid book name.
170 </div> -->
171
172 <!-- Error message for invalid or empty input -->
173 {% if error_message %}
174 <div class="alert alert-danger" id="errorMessage">
175 <strong>NOTE:</strong> {{ error_message }}
176 </div>
177 {% endif %}
178
179 <div class="row">
180 {% if data %}
181 {% for i in data %}
182 <div class="col-md-3">
183 <div class="book-card">
184 
185 <h4>{{i[0]}}</h4>
186 <p>{{i[1]}}</p>
187 </div>
188 </div>
189 {% endfor %}
190 {% endif %}
191 </div>
192 </div>
193
194 <!-- Script for validation and handling errors -->
195 <script>
196 document.getElementById('searchForm').addEventListener('submit', function(event) {

```

```

recommend.html 7 x
book-recommender-system > templates > recommend.html > html > head > style > .book-card h4 > .book-card p
2 <html lang="en">
141 <body>
195 <script>
196 document.getElementById('searchForm').addEventListener('submit', function(event) {
197     const userInput = document.getElementById('bookInput').value.trim();
198     if (!userInput) {
199         event.preventDefault();
200         document.getElementById('errorMessage').style.display = 'block';
201     }
202 });
203
204 // document.getElementById('clearBtn').addEventListener('click', function() {
205 //     // Reload the page to clear all fields and results
206 //     // location.reload();
207
208 //     document.getElementById('bookInput').value = ''; // Clear the input
209 //     document.getElementById('errorMessage').style.display = 'none'; // Hide error message
210
211 //     // Clear the displayed results
212 //     const rowDiv = document.querySelector('.row');
213 //     rowDiv.innerHTML = ''; // Clear all contents of the row
214
215 //     while (rowDiv.firstChild) {
216 //         rowDiv.removeChild(rowDiv.firstChild); // Remove all child elements (book cards)
217 //     }
218
219 //     // Optionally, reset the form to its initial state
220 //     // document.getElementById('searchForm').reset(); // This resets the form fields, if any
221
222 //     // Delay the reload for a brief moment to ensure everything is cleared
223 //     setTimeout(function() {
224 //         location.reload(); // Reload the page to clear all fields and results
225 //     }, 1); // Adjust the timeout duration if necessary
226
227 // });
228
229 document.getElementById('clearBtn').addEventListener('click', function() {

```

recommend.html 7 x

book-recommender-system > templates > recommend.html > html > head > style > .book-card h4 > .book-card p

```

2   <html lang="en">
141 <body>
195   <script>
228
229       document.getElementById('clearBtn').addEventListener('click', function() {
230           // Clear the input field
231           document.getElementById('bookInput').value = '';
232
233           // Hide the error message if it exists
234           const errorMessage = document.getElementById('errorMessage');
235           if (errorMessage) {
236               errorMessage.style.display = 'none'; // Hide error message
237           }
238
239           // Clear the displayed results
240           const rowDiv = document.querySelector('.row');
241           if (rowDiv) {
242               rowDiv.innerHTML = ''; // Clear all contents of the row
243           }
244       });
245
246
247       // If no data is found, display the 'noResultsMessage'
248       {% if not data %}
249       document.getElementById('noResultsMessage').style.display = 'block';
250       {% endif %}
251   </script>
252
253 </body>
254 </html>
255

```

Chapter 7

Output Snapshots

1) Initial view of Top Books:

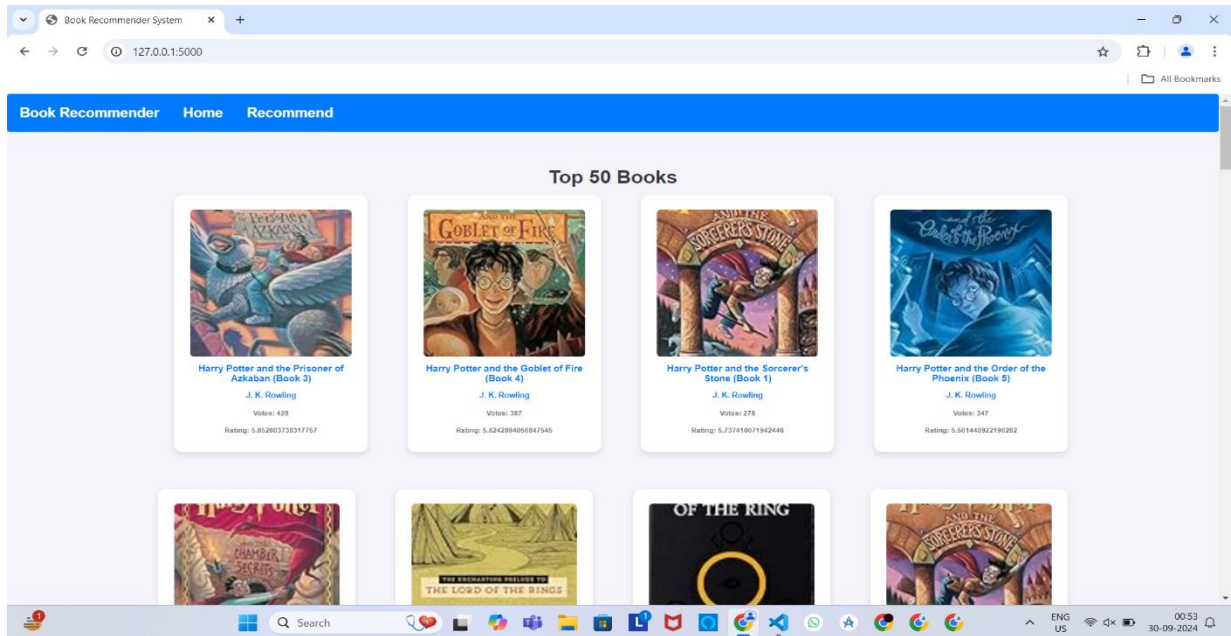


Fig: 7.1

2) Get recommendation page:

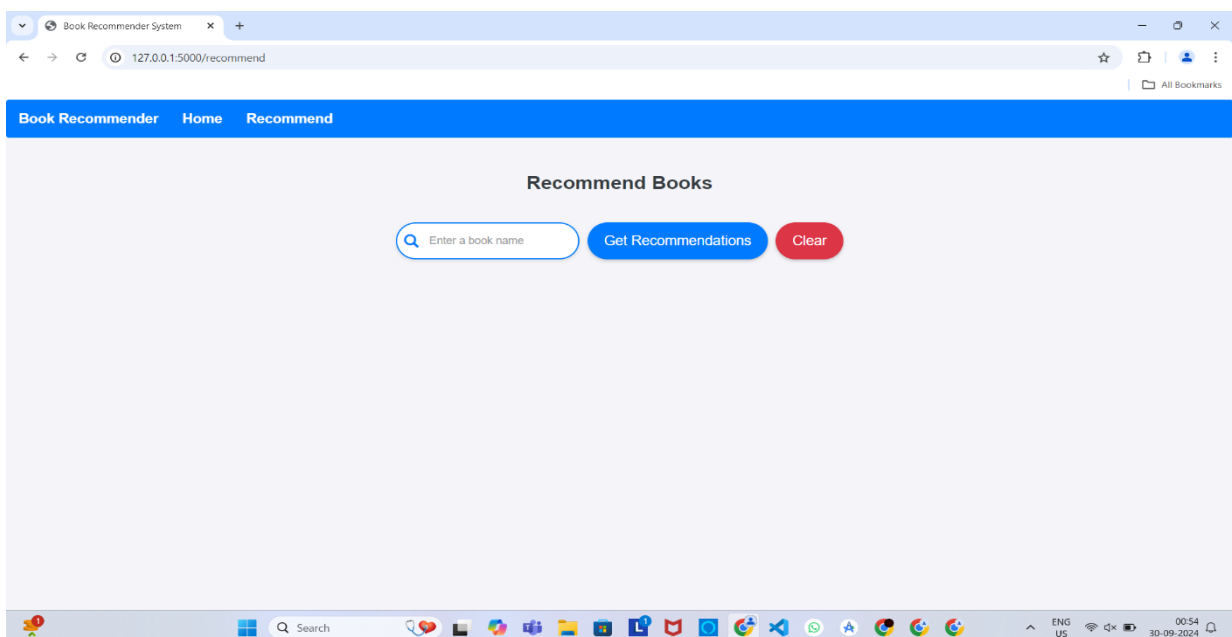


Fig: 7.2

3) Recommended results:

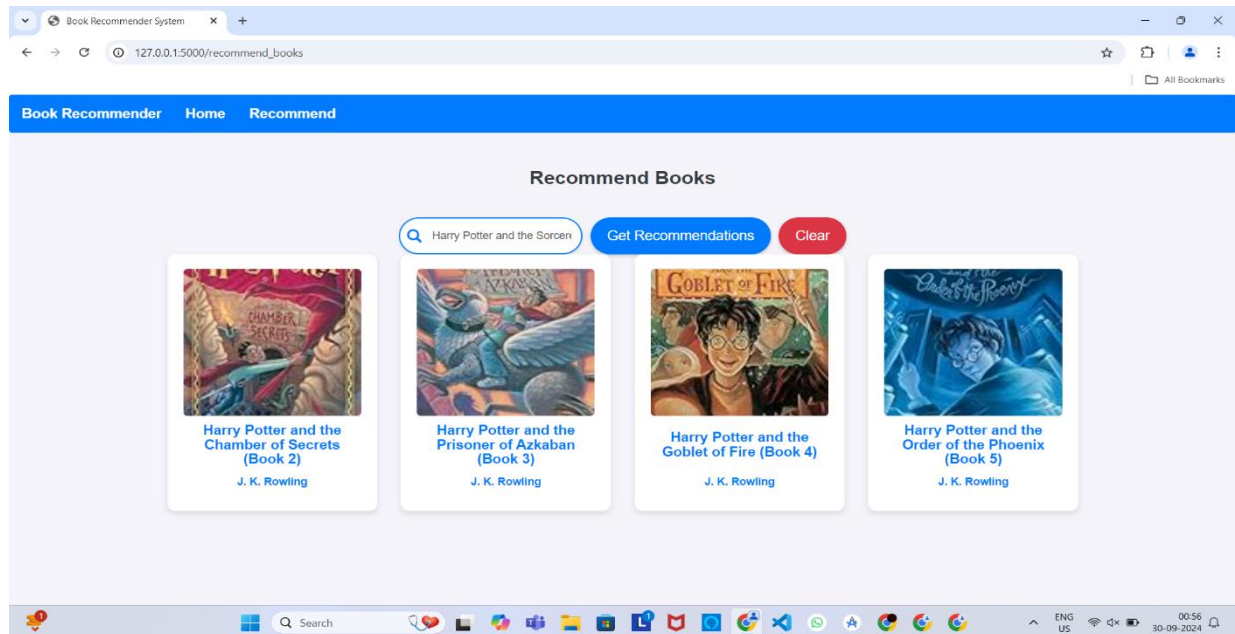


Fig: 7.3

4) Recommended results:

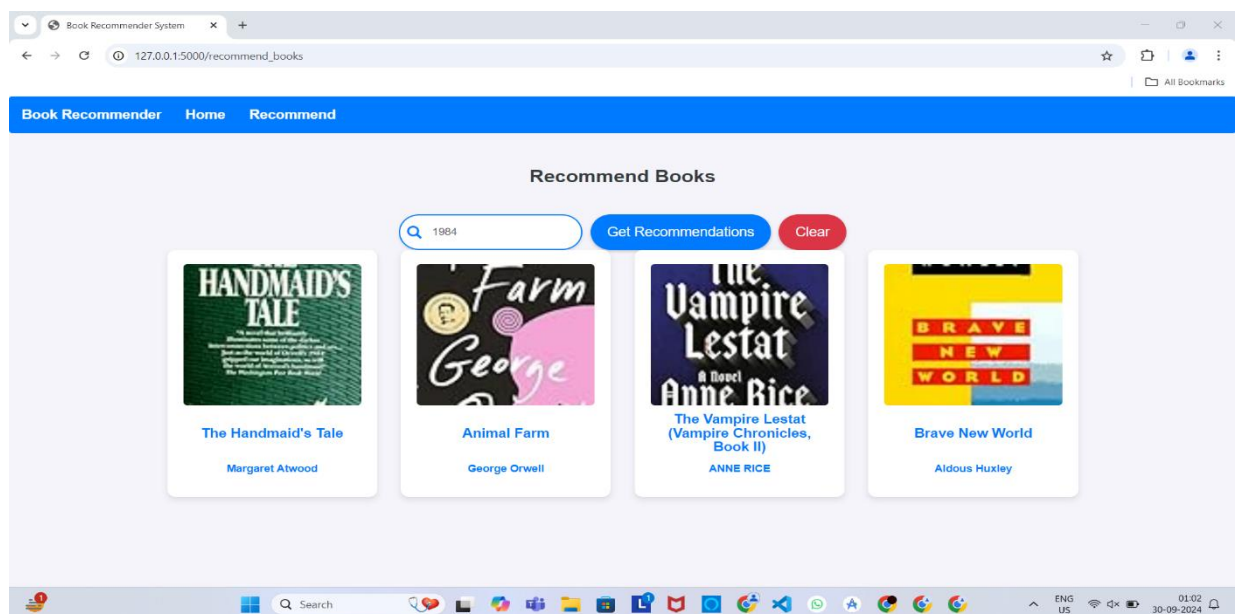


Fig: 7.4

Chapter 8

Conclusion

The **Book Recommendation System** effectively implements a dual recommendation strategy, combining popularity-based recommendations and collaborative filtering techniques. By utilizing **Machine Learning (ML) & Natural Language Processing (NLP)**, the system demonstrates the potential of modern recommendation systems to significantly improve user experiences on web platforms. The simplicity and intuitiveness of the user interface allow users to easily discover popular books or obtain personalized recommendations based on their specific input.

Key aspects of the system include:

➤ **User Interaction:**

The system enables seamless interaction, where users input book titles and receive tailored recommendations. Flask handles these interactions efficiently, ensuring the system remains responsive even under different user requests.

➤ **Efficient Computation:**

By precomputing models and leveraging serialized files for data storage, the system quickly processes large datasets and delivers recommendations without unnecessary delays, ensuring a smooth user experience.

➤ **Engaging User Experience:**

Balancing static, popularity-based suggestions with dynamic, collaborative filtering results, the system effectively caters to a wide range of user preferences, enhancing engagement and satisfaction.

The project showcases the integration of **Flask** for web application development and **Python** for machine learning, creating a scalable and robust solution. Additionally, the system provides opportunities for future extensions, such as adding user profiles, real-time recommendation updates, and advanced personalization techniques. Through a successful combination of web development and data science, this recommendation platform delivers both functional and personalized results, making it a powerful tool for book discovery and recommendation.

Chapter 9

Reference

1. <https://www.kaggle.com/datasets/arashnic/book-recommendation-dataset>
2. Kaggle
3. Youtube

THANK YOU !